

# Notes: Aguirregabiria and Magesan (RES, 2020)

## Identification and Estimation of Dynamic Games When Players' Beliefs Are Not in Equilibrium

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# 1 Big picture

- **Question.** How can we identify and estimate dynamic games when players are rational and forward-looking, but their beliefs about rivals' actions are *not* required to be in equilibrium?
- **Setting.** The paper studies dynamic discrete games of incomplete information in which players choose among finitely many actions, observe payoff-relevant states, and form possibly biased beliefs about rivals' current and future behavior.
- **Core challenge.** In standard empirical games, researchers often impose equilibrium beliefs so that beliefs are recovered directly from observed conditional choice probabilities (CCPs). Once beliefs are allowed to be biased, payoffs, beliefs, and continuation values become intertwined and standard identification breaks down.
- **Main methodological result.** A standard exclusion restriction in players' payoffs generates *testable restrictions* on the null hypothesis of equilibrium beliefs. The paper uses this idea to build a simple likelihood-ratio test for unbiased beliefs.
- **Second methodological result.** The same exclusion restriction, combined with knowledge of beliefs at only a subset of states, is sufficient to point identify contemporaneous beliefs and useful payoff objects even when beliefs outside that subset are biased.
- **Third methodological result.** The authors provide a practical estimation strategy: a nonparametric two-step estimator, plus a parametric extension based on pseudo maximum likelihood.
- **Why this matters.** The paper says researchers should *not* impose equilibrium beliefs by default. They should test them first, because imposing them when false can generate large structural bias.
- **Empirical application.** The method is applied to a dynamic game of store location by McDonalds and Burger King in the UK, 1991–1995.
- **Main empirical result.** The paper finds significant evidence of biased beliefs for Burger King: Burger King appears to underestimate McDonalds' probability of entry in geographically distant, relatively unexplored markets.
- **Economic takeaway.** Rational behavior does not imply equilibrium beliefs. In dynamic strategic settings, allowing beliefs to be biased can matter a lot for identification, estimation, and substantive conclusions about competition.

## 2 Model, data, and key objects

### 2.1 Dynamic game environment

- There are  $N$  players, periods  $t = 1, \dots, T$ , and a finite action set

$$Y = \{0, 1, \dots, A - 1\}.$$

- Player  $i$  chooses action  $Y_{it}$  to maximize expected discounted payoffs.

- The one-period payoff is

$$u_{it} = \pi_{it}(Y_{it}, Y_{-it}, X_t) + \varepsilon_{it}(Y_{it}),$$

where:

- $\pi_{it}(\cdot)$  is the systematic payoff component,
- $Y_{-it}$  are rivals' current actions,
- $X_t$  is the vector of common-knowledge state variables,
- $\varepsilon_{it}(Y_{it})$  is player  $i$ 's private payoff shock.
- The common state evolves according to

$$f_t(X_{t+1} \mid Y_t, X_t).$$

- Private shocks are independent across players and time and have known distribution up to scale.

## 2.2 Illustrative market-entry game

The paper's running example is a dynamic market entry-exit game. Firm  $i$  is active when  $Y_{it} = 1$ . If active, its profit is

$$\pi_{it}(1, Y_{-it}, X_t) = H_t \left( \theta_i^M - \theta_i^D \sum_{j \neq i} Y_{jt} \right) - \theta_{i0}^{FC} - \theta_{i1}^{FC} Z_{it} - \mathbf{1}\{Y_{i,t-1} = 0\} \theta_i^{EC}.$$

- $H_t$  is market size.
- $\theta_i^M$  is monopoly profit per capita.
- $\theta_i^D$  captures competition from rivals.
- $\theta_{i0}^{FC} + \theta_{i1}^{FC} Z_{it}$  is fixed cost.
- $\theta_i^{EC}$  is sunk entry cost when the firm was previously inactive.

*Interpretation.* This example makes clear what the paper wants to identify: structural payoffs, competitive effects, sunk costs, and players' beliefs about rivals' choices.

## 2.3 Strategies, CCPs, and beliefs

- A strategy for player  $i$  is  $\sigma_{it}(X_t, \varepsilon_{it})$ .
- The corresponding conditional choice probability (CCP) is

$$P_{it}(y \mid x) = \int \mathbf{1}\{\sigma_{it}(x, \varepsilon_{it}) = y\} dG_{it}(\varepsilon_{it}).$$

- When beliefs are not forced to be in equilibrium, player  $i$ 's belief at date  $t$  about rivals' actions at date  $t + s$  is

$$B_{i,t+s}^{(t)}(y_{-i} \mid x_{t+s}).$$

The key point is that beliefs may depend on the period when they are formed, so beliefs about period  $t + s$  formed at  $t$  need not equal beliefs about the same period formed later.

## 2.4 Behavioral assumptions

The paper keeps the following assumptions:

- **MOD-1:** strategies depend only on payoff-relevant states  $(X_t, \varepsilon_{it})$ ;
- **MOD-2:** players maximize expected discounted payoffs given beliefs;
- **MOD-3:** players have unbiased beliefs about their *own* future behavior;
- **MOD-4:** players know rivals' private shocks are independent across players, so beliefs over rivals' joint actions factorize;
- **MOD-5:** the common state space  $X$  is discrete and finite.

*Interpretation.* The paper relaxes equilibrium beliefs but keeps a revealed-preference notion of rationality.

## 2.5 Data structure for identification

- The researcher observes panel data across many markets  $m = 1, \dots, M$  and a limited number of periods  $t = 1, \dots, T_{\text{data}}$ .
- The key observables are actions  $Y_{imt}$  and state variables  $X_{mt}$ .
- Under Assumption ID-1, observed CCPs are identified from the data:

$$P_{it}^0(y \mid x) = \Pr(Y_{imt} = y \mid X_{mt} = x).$$

The paper assumes transition probabilities, the discount factor, and the distribution of private shocks are known or separately identified, so the hard objects are payoffs and beliefs.

## 2.6 Empirical application data

- The application uses a five-year panel of **422** UK local authority districts from **1991–1995**, excluding Greater London districts.
- The median district has area about **300 square km** and population about **94.85 thousand**.
- In 1990, Burger King (BK) had **79** stores in **71** markets; McDonalds (MD) had **281** stores in **206** markets.
- By 1995, BK had **222** stores in **150** markets, while MD had **447** stores in **254** markets.

*Interpretation.* The application is well suited to the paper because fast-food chains make forward-looking location decisions, entry is partly irreversible, and strategic uncertainty about rival expansion is plausible in the early UK market.

### 3 Models and empirical specifications (all equations + interpretation)

#### 3.1 Best response mapping

Given beliefs  $B_i^{(t)} = \{B_{ij,t+s}^{(t)} : s \geq 0\}$ , player  $i$  solves a dynamic programming problem. The expected one-period payoff is

$$\pi_{i,t+s}^{B^{(t)}}(y_{i,t+s}, x_{t+s}) = \sum_{y_{-i} \in Y^{N-1}} \pi_{i,t+s}(y_{i,t+s}, y_{-i}, x_{t+s}) B_{i,t+s}^{(t)}(y_{-i} \mid x_{t+s}),$$

and the induced transition is

$$f_{i,t+s}^{B^{(t)}}(x_{t+s+1} \mid y_{i,t+s}, x_{t+s}) = \sum_{y_{-i} \in Y^{N-1}} f_{t+s}(x_{t+s+1} \mid y_{i,t+s}, y_{-i}, x_{t+s}) B_{i,t+s}^{(t)}(y_{-i} \mid x_{t+s}).$$

The Bellman equation is

$$V_{it}^{B^{(t)}}(x_t, \varepsilon_{it}) = \max_{y_{it} \in Y} \left\{ v_{it}^{B^{(t)}}(y_{it}, x_t) + \varepsilon_{it}(y_{it}) \right\},$$

with conditional choice value

$$v_{it}^{B^{(t)}}(y_{it}, x_t) = \pi_{it}^{B^{(t)}}(y_{it}, x_t) + \beta \sum_{x_{t+1}} \left[ \int V_{i,t+1}^{B^{(t)}}(x_{t+1}, \varepsilon_{i,t+1}) dG_{it}(\varepsilon_{i,t+1}) \right] f_{it}^{B^{(t)}}(x_{t+1} \mid y_{it}, x_t).$$

This implies the best-response probability representation

$$P_{it}(y \mid x) = \Phi_{it}(y; \Delta v_{it}^{B^{(t)}}(x)).$$

If the shocks are i.i.d. extreme value type I, then CCPs have the logit form

$$P_{it}(y \mid x) = \frac{\exp\{\Delta v_{it}^{B^{(t)}}(y, x)\}}{\sum_{y' \in Y} \exp\{\Delta v_{it}^{B^{(t)}}(y', x)\}}.$$

*Interpretation.* Observed actions identify value differences, but those value differences depend on beliefs. This is the core reason beliefs matter for structural inference.

#### 3.2 Hotz–Miller inversion

The paper first uses the standard inversion theorem. If

$$P_{it}(x) = \Phi(\Delta v_{it}^{B^{(t)}}(x)),$$

then there is a one-to-one inverse map  $q(\cdot)$  such that

$$\Delta v_{it}^{B^{(t)}}(x) = q(P_{it}(x)).$$

In the logit case,

$$q(y, P_{it}(x)) = \ln P_{it}(y \mid x) - \ln P_{it}(0 \mid x).$$

*Interpretation.* CCPs identify value differences, but not beliefs and payoffs separately.

### 3.3 Why the model is underidentified without extra structure

Let  $q_{it}(y_i, x)$  denote the identified value difference from Hotz–Miller inversion. Then the paper writes

$$q_{it}(y_i, x) = B_{it}^{(t)'}(x) \left[ \pi_{it}(y_i, x) + \Delta c_{it}^{B^{(t)}}(y_i, x) \right]. \quad (3.1)$$

Here:

- $B_{it}^{(t)}(x)$  is the vector of contemporaneous beliefs,
- $\pi_{it}(y_i, x)$  is the vector of payoff terms across rival actions,
- $\Delta c_{it}^{B^{(t)}}(y_i, x)$  is the vector of continuation-value differences.

*Interpretation.* Even though  $q_{it}(y_i, x)$  is identified, this equation contains *three* unknown objects: current payoffs, contemporaneous beliefs, and continuation values. Without additional restrictions, the model is underidentified.

### 3.4 Exclusion restriction

The key extra structure is Assumption ID-3. The state vector is partitioned as

$$X_t = (S_t, W_t), \quad S_t = (S_{1t}, \dots, S_{Nt}),$$

where  $S_{it}$  enters player  $i$ 's payoff but is excluded from other players' payoffs:

$$\pi_{it}(Y_{it}, Y_{-it}, S_{it}, S_{-it}, W_t) = \pi_{it}(Y_{it}, Y_{-it}, S_{it}, S'_{-it}, W_t) \quad \text{for all } S'_{-it} \neq S_{-it}.$$

The exclusion restriction is complemented by:

- $|S| \geq A$ ,
- strictly positive support over  $(S_i, S_{-i}, W)$ ,
- a transition rule for  $S_{i,t+1}$  that depends on  $(Y_{it}, W_t)$  but not on  $(S_{it}, S_{-it})$  conditional on those variables.

*Interpretation.* Rival-specific lagged state variables can shift beliefs without directly changing the current payoff of player  $i$ . That is what lets the researcher separate beliefs from payoffs.

### 3.5 Test of equilibrium beliefs

Proposition 1 is the paper's first central result. In a two-player game, under the exclusion restriction, one can construct from player  $i$ 's own CCPs an object that depends only on player  $i$ 's beliefs. Under unbiased beliefs, the same object is identified from player  $j$ 's CCPs. Therefore, the null of equilibrium beliefs implies equality between the two.

In the binary-choice case, for a triple  $(s^0, s^{(a)}, s^{(b)})$ , the restriction becomes

$$\frac{q_{it}(1, s_i, s^{(a)}, w) - q_{it}(1, s_i, s^0, w)}{q_{it}(1, s_i, s^{(b)}, w) - q_{it}(1, s_i, s^0, w)} = \frac{P_{jt}(1 \mid s_i, s^{(a)}, w) - P_{jt}(1 \mid s_i, s^0, w)}{P_{jt}(1 \mid s_i, s^{(b)}, w) - P_{jt}(1 \mid s_i, s^0, w)}. \quad (3.4)$$

*Interpretation.* This is clever: the exclusion restriction strips out payoffs, leaving a testable relationship between own CCPs and rival CCPs that must hold if beliefs are unbiased.

The formal likelihood-ratio test compares:

- an unconstrained multinomial likelihood for the CCPs,
- a constrained likelihood imposing the equilibrium-beliefs restrictions.

The LR statistic is

$$LR = 2\left(\ell(\hat{P}_i^u, \hat{P}_j^u) - \ell(\hat{P}_i^c, \hat{P}_j^c)\right), \quad (4.4)$$

which is asymptotically chi-squared under the null.

### 3.6 Testing monotonicity of beliefs

The same logic also delivers a test for whether beliefs are monotone in an ordered player-specific state. In the binary case, define

$$\delta_{it}(s^{(1)}, s^{(2)}, s^{(3)}) = \frac{q_{it}(1, s_i, s^{(3)}, w) - q_{it}(1, s_i, s^{(2)}, w)}{q_{it}(1, s_i, s^{(2)}, w) - q_{it}(1, s_i, s^{(1)}, w)}. \quad (3.5)$$

The paper shows

$$\delta_{it}(s^{(1)}, s^{(2)}, s^{(3)}) = \frac{B_{ijt}^{(t)}(1, s_i, s^{(3)}, w) - B_{ijt}^{(t)}(1, s_i, s^{(2)}, w)}{B_{ijt}^{(t)}(1, s_i, s^{(2)}, w) - B_{ijt}^{(t)}(1, s_i, s^{(1)}, w)}. \quad (3.6)$$

So the sign of  $\delta_{it}$  reveals whether beliefs are monotone in  $S_j$ .

*Interpretation.* The paper does not only test “equilibrium or not.” It can also test qualitative structure in the belief function itself.

### 3.7 Identification with partially unbiased beliefs

The second central result is Assumption ID-4 and Proposition 2. Let  $S^{(R)} \subset S$  be a subset of states where the researcher knows beliefs, either because:

- beliefs are unbiased there, or
- elicited-belief data are available there.

If the associated matrix of rival CCPs over those states has sufficient rank, then:

- contemporaneous beliefs are identified everywhere,
- the function

$$g_{it}^B(y_i, y_{-i}, s_i, w) = \pi_{it}(y_i, y_{-i}, s_i, w) + \Delta c_{it}^B(y_i, y_{-i}, w)$$

is identified everywhere,

- payoff *differences* across values of the player-specific state are identified.



For general  $N$  and  $A$ , the order condition is

$$\frac{1 - (R/|S|)}{1 - (A/|S|)^{N-1}} \leq \frac{1}{N-1}. \quad (3.9)$$

For the two-player case, this reduces to

$$R \geq A.$$

*Interpretation.* The method does *not* need unbiased beliefs everywhere. It only needs a small enough anchor set of states where beliefs are known.

### 3.8 Where should one assume unbiased beliefs?

The paper discusses three practical rules for choosing the set  $S^{(R)}$ :

1. Use the equilibrium-beliefs test and pick states from triples that are not rejected.
2. If the player-specific variable is ordered, test for monotonicity and use extreme points where unbiased beliefs are plausible.
3. Choose the set that minimizes estimated belief bias.

*Interpretation.* The paper is unusually practical here: it does not just prove identification; it discusses how an applied researcher can choose the anchor states.

### 3.9 Nonparametric estimation

The paper's nonparametric estimator has two main steps:

**Step 1.** Estimate CCPs and transition probabilities nonparametrically; invert CCPs into value differences.

**Step 2.** Given the chosen set  $S^{(R)}$ , recover:

- the  $g^B$  object by

$$g_{it}^B(y_i, s_i, w) = (P_{-it}^{(R)'}(s_i, w) P_{-it}^{(R)}(s_i, w))^{-1} P_{-it}^{(R)'}(s_i, w) q_{it}^{(R)}(y_i, s_i, w), \quad (4.5)$$

- contemporaneous beliefs by matrix inversion

$$B_{it}^{(t)}(x) = [\tilde{G}_{it}(x)]^{-1} q_{it}(x), \quad (4.6)$$

- payoff levels from identified payoff differences:

$$\pi_{it}(y_i, y_j, s_i, w) - \pi_{it}(y_i, y_j, s^*, w) = g_{it}^B(y_i, y_j, s_i, w) - g_{it}^B(y_i, y_j, s^*, w). \quad (4.7)$$

*Interpretation.* The estimator is constructive: each identification proof becomes an explicit estimation formula.

### 3.10 Parametric payoff extension

If payoffs are linear in parameters,

$$\pi_{it}(Y_{it}, Y_{jt}, S_{it}, W_t) = h(Y_{it}, Y_{jt}, S_{it}, W_t)\theta_{it}, \quad (4.8)$$

then the paper adds a third step:

- estimate beliefs and CCPs as above,
- then estimate payoff parameters by pseudo maximum likelihood.

The pseudo-likelihood has the form

$$Q(\theta_i, B_i, P_i) = \sum_{m=1}^M \sum_{t=1}^{T_{\text{data}}} \log \left( \frac{\exp\{\tilde{h}_{it}^{B,P}(y_{imt}, x_{mt})\theta_i + \tilde{e}_{it}^{B,P}(y_{imt}, x_{mt})\}}{\sum_{y_i=0}^{A-1} \exp\{\tilde{h}_{it}^{B,P}(y_i, x_{mt})\theta_i + \tilde{e}_{it}^{B,P}(y_i, x_{mt})\}} \right). \quad (4.10)$$

*Interpretation.* This makes the framework usable in applications where the researcher wants structural parameters, not only nonparametric payoff objects.

### 3.11 Monte Carlo design

The Monte Carlo section studies a two-player binary entry game. The payoffs are

$$\begin{aligned} \pi_{1mt}(1, Y_{2mt}, X_{mt}) &= (1 - Y_{2mt})\theta_1^M + Y_{2mt}\theta_1^D - \theta_{01}^{FC} - (1 - Y_{1m,t-1})\theta_1^{EC}, \\ \pi_{2mt}(1, Y_{1mt}, X_{mt}) &= (1 - Y_{1mt})\theta_2^M + Y_{1mt}\theta_2^D - \theta_{02}^{FC} - \theta_{12}^{FC} Z_{2mt} - (1 - Y_{2m,t-1})\theta_2^{EC}. \end{aligned} \quad (5.1)$$

The key excluded variable is  $Z_{2mt}$ , which enters player 2's payoff but not player 1's directly.

There are two experiments:

- **Experiment U:** beliefs are in equilibrium.
- **Experiment B:** beliefs are biased, with

$$\lambda_1 = \lambda_2 = 1 \quad \text{if } Z_2 \in \{-2, 2\}, \quad \lambda_1 = \lambda_2 = 0.5 \quad \text{if } Z_2 \in \{-1, 0, 1\}. \quad (5.3)$$

*Interpretation.* In the biased-beliefs experiment, players systematically underpredict rivals' activity in interior states, so the paper can test how dangerous wrong equilibrium restrictions are.

### 3.12 Empirical application model: McDonalds and Burger King

In the application, firms choose whether to open a new store in a district. Let  $K_{imt}$  denote the stock of stores of firm  $i$  in market  $m$  at  $t - 1$  and  $Y_{imt} \in \{0, 1\}$  the number of new stores opened at  $t$ .

Variable profits are

$$VP_{imt} = (W_m \gamma) N_{imt} (\theta_{0i}^{VP} + \theta_{can,i}^{VP} N_{imt} + \theta_{com,i}^{VP} N_{jmt}), \quad (6.1)$$

where  $N_{imt} = K_{imt} + Y_{imt}$ .

Entry costs are

$$EC_{imt} = \mathbf{1}\{Y_{imt} > 0\}(\theta_{0i}^{EC} + \theta_{K,i}^{EC}\mathbf{1}\{K_{imt} > 0\} + \theta_{Z,i}^{EC}Z_{imt} + \varepsilon_{it}), \quad (6.2)$$

and fixed operating costs are

$$FC_{imt} = \mathbf{1}\{N_{imt} > 0\}(\theta_{0i}^{FC} + \theta_{lin,i}^{FC}N_{imt} + \theta_{qua,i}^{FC}N_{imt}^2). \quad (6.3)$$

- $W_m$  contains district characteristics such as population, density, age structure, GDP per capita, rent, and unemployment.
- $\theta_{can,i}^{VP}$  captures own-store cannibalization or economies of scope.
- $\theta_{com,i}^{VP}$  captures competition from the rival chain.
- $\theta_{Z,i}^{EC}$  captures economies of density through distance to the chain's existing network.

The key empirical exclusion restriction is that the rival's lagged network affects beliefs but not current profits directly once current rival stores are conditioned on.

*Interpretation.* This is where the method becomes operational: rival lagged network variables are the excluded belief shifters.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What fails in the standard approach

- In standard empirical games, one typically sets beliefs equal to observed CCPs.
- If beliefs are biased, this is a misspecification.
- The paper shows that with biased beliefs, imposing equilibrium restrictions can severely bias competition estimates.

*Main point.* “Rational” need not mean “equilibrium beliefs.”

### 4.2 What identifies beliefs

The identification logic has three pieces:

1. Hotz–Miller inversion identifies value differences from CCPs.
2. An exclusion restriction creates state variation that shifts beliefs without directly shifting payoffs.
3. A subset of states where beliefs are known anchors the full belief function.

*Economic intuition.* If a rival-specific lagged state affects my action only through what I believe about the rival, then my action reveals those beliefs.

### 4.3 Why the exclusion restriction is plausible in the application

For the McDonalds–Burger King game:

- a firm’s *own* lagged store network affects current profits through sunk entry costs and economies of density;
- the rival’s lagged network does not directly affect current profits, once current rival stores are controlled for;
- but the rival’s lagged network should matter for beliefs about where the rival may enter next.

*Interpretation.* Rival lagged geography is an “instrument-like” source of variation for beliefs.

### 4.4 Why partially unbiased beliefs are plausible

The paper assumes beliefs are unbiased only in markets geographically close to the rival’s existing network:

$$B_{imt}(y_j \mid x_{mt}) = P_{jmt}(y_j \mid x_{mt}) \quad \text{if } z_{jmt} \leq z^*. \quad (6.4)$$

- Close to the rival’s network, the rival’s strategy is easier to infer from past experience.
- In distant, less explored markets, strategic uncertainty is more plausible.

This is exactly the kind of structured but weak assumption the identification theory needs.

### 4.5 Choice of the anchor set $S^{(R)}$ in the application

The authors use several considerations:

- BK and MD entry probabilities are strictly decreasing in their own distance variable  $Z$ .
- Beliefs are not rejected as monotone in this variable.
- Estimated models with unbiased beliefs imposed at small  $Z$  values are more conservative and imply smaller belief distortions than models imposing unbiased beliefs at large  $Z$  values.

*Interpretation.* They do not pick the anchor states arbitrarily; they use both theory and data diagnostics.

### 4.6 Empirical credibility and limitations

- The UK fast-food hamburger market in the early 1990s was still expanding, making dynamic entry incentives natural.
- Store openings are effectively irreversible during the sample.
- Greater London is excluded because it does not fit the isolated-local-market assumption.

A limitation is that the empirical application is only two players and a relatively short panel, but that is also what makes it a clean illustration of the method.

## 4.7 Relation to the literature

- The paper relaxes the Markov-perfect-equilibrium-beliefs assumption standard in empirical dynamic games.
- It differs from rationalizability approaches that rely on level- $k$  reasoning bounds.
- It is also related to the literature using elicited beliefs, but here beliefs are recovered structurally without needing beliefs at every state.

## 5 Tables and figures

### 5.1 Table 1 and Table 2: why exclusion restrictions matter

TABLE 1  
Order condition for identification models without exclusion restrictions in payoffs (number of parameters & restrictions for each player-period)

| Number of parameters and restrictions    | Models without exclusion restrictions    |                                  |
|------------------------------------------|------------------------------------------|----------------------------------|
|                                          | (A) Unrestricted beliefs                 | (B) Unbiased (Equil) beliefs     |
| (1) Restrictions from observed behaviour | $(A-1)  \mathcal{X} $                    | $(A-1)  \mathcal{X} $            |
| (2) Restrictions from unbiased beliefs   | 0                                        | $(N-1)(A-1)  \mathcal{X} $       |
| (3) Free parameters in payoffs           | $(A-1)  \mathcal{X}  A^{N-1}$            | $(A-1)  \mathcal{X}  A^{N-1}$    |
| (4) Free parameters in beliefs           | $(N-1)(A-1)  \mathcal{X} $               | $(N-1)(A-1)  \mathcal{X} $       |
| (5) Free parameters in $\tilde{c}$       | $(A-1)  \mathcal{X} $                    | $(A-1)  \mathcal{X} $            |
| (1)+(2)–(3)–(4)–(5)                      | $-(A-1)  \mathcal{X}  [A^{N-1} + (N-1)]$ | $-(A-1)  \mathcal{X}  [A^{N-1}]$ |
| Over-under identifying rest              | NO                                       | NO                               |
| Is the model identified?                 | NO                                       | NO                               |

TABLE 2  
Order condition for identification models WITH exclusion restrictions in payoffs (number of parameters and restrictions for each player-period)

| Number of parameters and restrictions    | Models with exclusion restrictions                                                   |                                                            |
|------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------|
|                                          | (A) Unrestricted beliefs                                                             | (B) Unbiased (Equil) beliefs                               |
| (1) Restrictions from observed behaviour | $(A-1)  S ^N$                                                                        | $(A-1)  S ^N$                                              |
| (2) Restrictions from unbiased beliefs   | 0                                                                                    | $(N-1)(A-1)  S ^N$                                         |
| (3) Free parameters in payoffs           | $(A-1)  S  A^{N-1}$                                                                  | $(A-1)  S  A^{N-1}$                                        |
| (4) Free parameters in beliefs           | $(N-1)(A-1)  S ^N$                                                                   | $(N-1)(A-1)  S ^N$                                         |
| (5) Free parameters in $\tilde{c}$       | $(A-1)$                                                                              | 0                                                          |
| (1)+(2)–(3)–(4)–(5)                      | $(A-1)  S ^N \left[ 1 - \frac{A^{N-1}}{ S ^{N-1}} - (N-1) - \frac{1}{ S ^N} \right]$ | $(A-1)  S ^N \left[ 1 - \frac{A^{N-1}}{ S ^{N-1}} \right]$ |
| Over-under identifying rest              | NO (For any $N \geq 2$ )                                                             | YES (For any $ S  \geq A$ )                                |
| Is the model identified?                 | NO                                                                                   | YES                                                        |

Read:

- Table 1 shows the basic problem: without exclusion restrictions, the model is underidentified both when beliefs are unrestricted and when beliefs are forced to be in equilibrium.
- Table 2 shows what exclusion restrictions buy. With exclusion restrictions, equilibrium beliefs overidentify the model, which is exactly what makes testing possible.
- But exclusion restrictions *alone* are still not enough when beliefs are fully unrestricted; the paper needs the weaker “partially unbiased beliefs” anchor.
- The big message is that the exclusion restriction is doing two jobs: it creates a test for equilibrium beliefs and it helps recover beliefs and payoffs once a subset of states is anchored.

### 5.2 Figures 1 and 2: power and size of the LR test

Read:

- Figure 1 studies the case where the null is false. The rejection probability is essentially one for standard significance levels, so the test has strong power against biased beliefs.
- Figure 2 studies the case where the null is true. The empirical rejection frequency tracks the 45° line closely, so the test has approximately correct size.
- Together, the two figures justify the paper’s empirical advice: test equilibrium beliefs before imposing them.

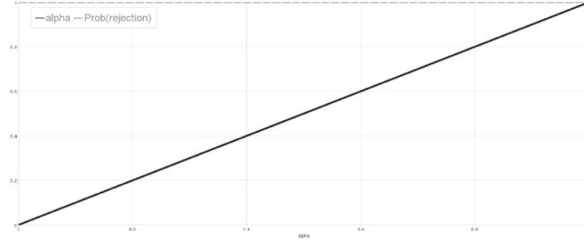


FIGURE 1  
Rejection probability when the null is false (Experiment "B")

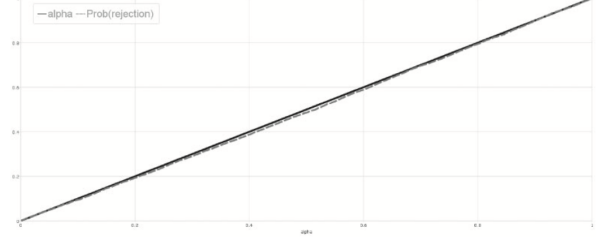


FIGURE 2  
Rejection probability when the null is true (Experiment "U")

### 5.3 Table 3 and Table 4: the bias–variance trade-off

TABLE 3  
Monte Carlo experiments<sup>(1)</sup>

| Parameter<br>(true value)                   | Biased beliefs in the DGP                      |                   |                                              |                   | Unbiased beliefs in the DGP                    |                   |                                              |                   |
|---------------------------------------------|------------------------------------------------|-------------------|----------------------------------------------|-------------------|------------------------------------------------|-------------------|----------------------------------------------|-------------------|
|                                             | Estimation<br>with equilibrium<br>restrictions |                   | Estimation<br>no equilibrium<br>restrictions |                   | Estimation<br>with equilibrium<br>restrictions |                   | Estimation<br>no equilibrium<br>restrictions |                   |
|                                             | (1)<br>MAB <sup>(2)</sup><br>(%)               | (2)<br>Std<br>(%) | (3)<br>MAB<br>(%)                            | (4)<br>Std<br>(%) | (5)<br>MAB<br>(%)                              | (6)<br>Std<br>(%) | (7)<br>MAB<br>(%)                            | (8)<br>Std<br>(%) |
| Payoffs                                     |                                                |                   |                                              |                   |                                                |                   |                                              |                   |
| $\alpha_1(2.4)$                             | 0.521<br>(21.7%)                               | 0.332<br>(13.8%)  | 0.308<br>(12.8%)                             | 0.245<br>(10.2%)  | $\alpha_1(2.4)$                                | 0.180<br>(7.5%)   | 0.135<br>(5.6%)                              | 0.525<br>(21.9%)  |
| $\delta_1(3.0)$                             | 0.456<br>(15.2%)                               | 0.277<br>(9.2%)   | 0.284<br>(9.5%)                              | 0.224<br>(7.5%)   | $\delta_1(3.0)$                                | 0.178<br>(5.9%)   | 0.134<br>(4.5%)                              | 0.581<br>(19.4%)  |
| $\theta_1^{EC}(0.5)$                        | 0.108<br>(21.6%)                               | 0.081<br>(16.3%)  | 0.205<br>(40.9%)                             | 0.160<br>(32.0%)  | $\theta_1^{EC}(0.5)$                           | 0.091<br>(18.3%)  | 0.069<br>(13.8%)                             | 0.172<br>(34.5%)  |
| Beliefs at $Z_2=0: B_1(Y_{1,-1}, Y_{2,-1})$ |                                                |                   |                                              |                   |                                                |                   |                                              |                   |
| $B_1(0,0)(0.410)$                           | 0.409<br>(99.8%)                               | 0.066<br>(16.0%)  | 0.148<br>(36.2%)                             | 0.110<br>(26.8%)  | $B_1(0,0)(0.658)$                              | 0.058<br>(8.8%)   | 0.044<br>(6.6%)                              | 0.101<br>(15.4%)  |
| $B_1(0,1)(0.442)$                           | 0.442<br>(100%)                                | 0.035<br>(7.9%)   | 0.111<br>(25.1%)                             | 0.086<br>(19.5%)  | $B_1(0,1)(0.814)$                              | 0.031<br>(3.8%)   | 0.023<br>(2.8%)                              | 0.081<br>(9.9%)   |
| $B_1(1,0)(0.403)$                           | 0.403<br>(100%)                                | 0.035<br>(8.7%)   | 0.091<br>(22.7%)                             | 0.072<br>(17.9%)  | $B_1(1,0)(0.559)$                              | 0.031<br>(5.6%)   | 0.024<br>(4.3%)                              | 0.068<br>(12.2%)  |
| $B_1(1,1)(0.437)$                           | 0.437<br>(100%)                                | 0.021<br>(4.8%)   | 0.070<br>(16.0%)                             | 0.053<br>(12.2%)  | $B_1(1,1)(0.727)$                              | 0.026<br>(3.6%)   | 0.020<br>(2.7%)                              | 0.054<br>(7.4%)   |
| CCPs at $Z_2=0: P_1(Y_{1,-1}, Y_{2,-1})$    |                                                |                   |                                              |                   |                                                |                   |                                              |                   |
| $P_1(0,0)(0.829)$                           | 0.073<br>(8.8%)                                | 0.034<br>(9.7%)   | 0.051<br>(4.1%)                              | 0.038<br>(4.6%)   | $P_1(0,0)(0.704)$                              | 0.037<br>(5.2%)   | 0.028<br>(4.0%)                              | 0.063<br>(8.9%)   |
| $P_1(0,1)(0.814)$                           | 0.090<br>(11.1%)                               | 0.026<br>(3.1%)   | 0.035<br>(4.3%)                              | 0.026<br>(3.2%)   | $P_1(0,1)(0.598)$                              | 0.025<br>(4.2%)   | 0.019<br>(3.2%)                              | 0.046<br>(7.6%)   |
| $P_1(1,0)(0.891)$                           | 0.056<br>(6.3%)                                | 0.015<br>(1.7%)   | 0.022<br>(2.5%)                              | 0.027<br>(3.1%)   | $P_1(1,0)(0.841)$                              | 0.014<br>(1.9%)   | 0.011<br>(1.3%)                              | 0.028<br>(3.3%)   |
| $P_1(1,1)(0.880)$                           | 0.072<br>(8.1%)                                | 0.013<br>(1.4%)   | 0.017<br>(1.9%)                              | 0.012<br>(1.4%)   | $P_1(1,1)(0.761)$                              | 0.016<br>(2.1%)   | 0.012<br>(1.6%)                              | 0.018<br>(3.1%)   |

Note (1): Summary of DGPs in Monte Carlo experiments. In all the experiments the values of the parameters are  $\alpha_1 = \alpha_2 = 2.4$ ,  $\delta_1 = \delta_2 = 3.0$ ,  $\theta_1^{EC} = -1.0$ ,  $\theta_2^{EC} = 0.5$ , and  $\beta_1 = \beta_2 = 0.95$ .  $Z_{2m} \sim \text{Uniform over } \{-2, -1, 0, +1, +2\}$  i.i.d. In Experiment B with biased beliefs,  $\lambda_1 = \lambda_2 = 1$  if  $Z_2 \in \{-2, 2\}$  and  $\lambda_1 = \lambda_2 = 0.5$  if  $Z_2 \in \{-1, 0, 1\}$ . Number of Monte Carlo Replications is 10,000. And number of observations in each replication,  $M = 500$  and  $T = 5$ . Note (2): MAB = Mean absolute bias. Std = Standard deviation. Note (3): (%) = Percentage of the true value of the parameter.

TABLE 4  
Monte Carlo experiment<sup>(1)</sup>: one replication with 1 million observations

| Parameter<br>(true value)                   | Biased Beliefs in the DGP                      |                   | Unbiased beliefs in the DGP                    |                   |
|---------------------------------------------|------------------------------------------------|-------------------|------------------------------------------------|-------------------|
|                                             | Estimation<br>with equilibrium<br>restrictions |                   | Estimation<br>with equilibrium<br>restrictions |                   |
|                                             | (1)<br>MAB <sup>(2)</sup><br>(%)               | (2)<br>MAB<br>(%) | (3)<br>MAB<br>(%)                              | (4)<br>MAB<br>(%) |
| Payoffs                                     |                                                |                   |                                                |                   |
| $\alpha_1(2.4)$                             | 0.523<br>(21.7%)                               | 0.005<br>(0.2%)   | $\alpha_1(2.4)$                                | 0.002<br>(0.1%)   |
| $\delta_1(3.0)$                             | 0.365<br>(12.2%)                               | 0.009<br>(0.3%)   | $\delta_1(3.0)$                                | 0.001<br>(0.0%)   |
| $\theta_1^{EC}(0.5)$                        | 0.051<br>(10.2%)                               | 0.003<br>(0.6%)   | $\theta_1^{EC}(0.5)$                           | 0.002<br>(0.3%)   |
| Beliefs at $Z_2=0: B_1(Y_{1,-1}, Y_{2,-1})$ |                                                |                   |                                                |                   |
| $B_1(0,0)(0.410)$                           | 0.408<br>(99.6%)                               | 0.007<br>(1.7%)   | $B_1(0,0)(0.658)$                              | 0.003<br>(0.4%)   |
| $B_1(0,1)(0.442)$                           | 0.444<br>(100%)                                | 0.007<br>(1.6%)   | $B_1(0,1)(0.814)$                              | 0.001<br>(0.1%)   |
| $B_1(1,0)(0.403)$                           | 0.399<br>(99.0%)                               | 0.004<br>(1.0%)   | $B_1(1,0)(0.559)$                              | 0.004<br>(0.6%)   |
| $B_1(1,1)(0.437)$                           | 0.437<br>(100%)                                | 0.006<br>(1.5%)   | $B_1(1,1)(0.727)$                              | 0.001<br>(0.1%)   |
| CCPs at $Z_2=0: P_1(Y_{1,-1}, Y_{2,-1})$    |                                                |                   |                                                |                   |
| $P_1(0,0)(0.829)$                           | 0.072<br>(8.7%)                                | 0.003<br>(0.4%)   | $P_1(0,0)(0.704)$                              | 0.001<br>(0.2%)   |
| $P_1(0,1)(0.814)$                           | 0.092<br>(11.3%)                               | 0.003<br>(0.4%)   | $P_1(0,1)(0.598)$                              | 0.001<br>(0.2%)   |
| $P_1(1,0)(0.891)$                           | 0.055<br>(6.3%)                                | 0.002<br>(0.2%)   | $P_1(1,0)(0.841)$                              | 0.001<br>(0.1%)   |
| $P_1(1,1)(0.880)$                           | 0.077<br>(8.2%)                                | 0.001<br>(0.2%)   | $P_1(1,1)(0.761)$                              | 0.001<br>(0.2%)   |

Note (1): Summary of DGPs in Monte Carlo Experiments. In all the experiments the values of the parameters are  $\alpha_1 = \alpha_2 = 2.4$ ,  $\delta_1 = \delta_2 = 3.0$ ,  $\theta_1^{EC} = -1.0$ ,  $\theta_2^{EC} = 0.5$ , and  $\beta_1 = \beta_2 = 0.95$ .  $Z_{2m} \sim \text{Uniform over } \{-2, -1, 0, +1, +2\}$  i.i.d. In Experiment B with Biased beliefs,  $\lambda_1 = \lambda_2 = 1$  if  $Z_2 \in \{-2, 2\}$  and  $\lambda_1 = \lambda_2 = 0.5$  if  $Z_2 \in \{-1, 0, 1\}$ . Number of Monte Carlo Replications is 1, and the number of observations is 1,000,000. Note (2): MAB = Mean absolute bias. Note (3): (%) = Percentage of the true value of the parameter.

#### Read:

- These tables quantify the core trade-off in the paper.
- If the data are generated by equilibrium beliefs, refusing to impose equilibrium restrictions reduces precision and increases finite-sample noise.
- But if the data are generated by biased beliefs, *imposing* equilibrium restrictions generates substantial bias.
- In the large-sample experiment with biased beliefs, the estimator that imposes equilibrium beliefs still has about **21.7%** bias for  $\alpha_1$ , **12.2%** bias for the competition parameter  $\delta_1$ , and **10.2%** bias for the entry-cost parameter, whereas the estimator allowing biased beliefs has near-zero bias.
- This is the central practical lesson of the Monte Carlo section: wrong equilibrium restrictions can be much more damaging than the precision gains they deliver.

### 5.4 Table 5 and Table 6: local markets and chain expansion

#### Read:

TABLE 5  
Descriptive statistics on local markets (Year 1991) (422 local authority districts (excluding Greater London districts))

| Variable                             | Median | Std. Dev. | Pctile 5% | Pctile 95% |
|--------------------------------------|--------|-----------|-----------|------------|
| Area (square km)                     | 300    | 732       | 36        | 1671       |
| Population (thousands)               | 94.85  | 93.04     | 37.10     | 280.50     |
| Share of children: Age 5–14 (%)      | 12.43  | 1.00      | 10.74     | 14.07      |
| Share of young: 15–29 (%)            | 21.24  | 2.46      | 17.80     | 25.17      |
| Share of pensioners: 65–74 (%)       | 9.01   | 1.50      | 6.89      | 11.82      |
| GDP per capita (thousand £)          | 92.00  | 12.14     | 74.40     | 112.70     |
| Claimants of UB/Population ratio (%) | 2.75   | 1.27      | 1.24      | 5.11       |
| Average weekly rent per dwelling (£) | 25.31  | 10.61     | 19.11     | 35.07      |
| Council tax (£)                      | 240    | 56        | 118       | 310        |
| Number of BK stores (Year 1990)      | 0.00   | 0.43      | 0.00      | 1.00       |
| Number of MD stores (Year 1990)      | 0.00   | 0.94      | 0.00      | 2.00       |

TABLE 6  
Evolution of the number of stores (422 local authority districts (excluding Greater London districts))

|                                                       | 1990 | 1991 | Burger King<br>1992 | 1993 | 1994 | 1995 |
|-------------------------------------------------------|------|------|---------------------|------|------|------|
| #Markets with stores                                  | 71   | 98   | 104                 | 118  | 131  | 150  |
| Change in #Markets with stores                        | –    | 27   | 6                   | 14   | 13   | 19   |
| # of stores                                           | 79   | 115  | 128                 | 153  | 181  | 222  |
| Change in # of stores                                 | –    | 36   | 13                  | 25   | 28   | 41   |
| Mean #stores per market<br>(conditional on #stores>0) | 1.11 | 1.17 | 1.23                | 1.30 | 1.38 | 1.48 |
|                                                       |      |      | McDonalds           |      |      |      |
| #Markets with stores                                  | 206  | 213  | 220                 | 237  | 248  | 254  |
| Change in #markets with stores                        | –    | 7    | 7                   | 17   | 11   | 6    |
| # of stores                                           | 281  | 316  | 344                 | 382  | 421  | 447  |
| Change in # of stores                                 | –    | 35   | 28                  | 38   | 39   | 26   |
| Mean #stores per market<br>(conditional on #stores>0) | 1.36 | 1.48 | 1.56                | 1.61 | 1.70 | 1.76 |

- Table 5 shows the local-market environment: median district population is about 95,000, and median area is about 300 square km.
- Table 6 shows that McDonalds is much larger than Burger King throughout the sample.
- BK grows from **79** stores in 1990 to **222** in 1995, while MD grows from **281** to **447**.
- Even though BK enters more new markets, MD still opens more stores overall. That already suggests asymmetry in expansion incentives and/or beliefs.

## 5.5 Table 7 and Table 8: transition patterns and reduced-form evidence

TABLE 7  
Transition probability matrix for market structure (Annual transitions. Market structure: BK = x & MD = y, where x and y are number of stores)

| Market<br>structure at $t$ | %                         |                  |                       |                  |                  |                       |                       |                       |                            |
|----------------------------|---------------------------|------------------|-----------------------|------------------|------------------|-----------------------|-----------------------|-----------------------|----------------------------|
|                            | Market structure at $t+1$ |                  |                       |                  |                  |                       |                       |                       |                            |
|                            | BK = 0<br>MD = 0          | BK = 0<br>MD = 1 | BK = 0<br>MD $\geq 2$ | BK = 1<br>MD = 0 | BK = 1<br>MD = 1 | BK = 1<br>MD $\geq 2$ | BK $\geq 2$<br>MD = 0 | BK $\geq 2$<br>MD = 1 | BK $\geq 2$<br>MD $\geq 2$ |
| BK = 0 & MD = 0            | 95.1                      | 3.6              | 0.2                   | 1.0              | —                | —                     | —                     | 0.1                   | —                          |
| BK = 0 & MD = 1            | —                         | 87.2             | 4.2                   | —                | 7.4              | 1.0                   | —                     | —                     | 1.4                        |
| BK = 0 & MD $\geq 2$       | —                         | —                | 82.7                  | —                | —                | 15.8                  | —                     | —                     | 1.4                        |
| BK = 1 & MD = 0            | —                         | —                | —                     | 76.0             | 18.0             | 2.0                   | 4.0                   | —                     | —                          |
| BK = 1 & MD = 1            | —                         | —                | —                     | —                | 87.1             | 8.1                   | —                     | 3.3                   | 1.4                        |
| BK = 1 & MD $\geq 2$       | —                         | —                | —                     | —                | —                | 86.5                  | —                     | —                     | 13.5                       |
| BK $\geq 2$ & MD = 0       | —                         | —                | —                     | —                | —                | —                     | 84.6                  | 15.4                  | —                          |
| BK $\geq 2$ & MD = 1       | —                         | —                | —                     | —                | —                | —                     | —                     | 69.0                  | 31.0                       |
| BK $\geq 2$ & MD $\geq 2$  | —                         | —                | —                     | —                | —                | —                     | —                     | —                     | 100.0                      |
| Frequency                  | 41.6                      | 23.3             | 6.6                   | 2.2              | 10.9             | 8.8                   | 0.6                   | 1.4                   | 4.5                        |

TABLE 8  
Reduced form probits for the decision to open a store

| Explanatory variable                   | Estimated marginal effects <sup>1</sup> (ΔP(x) when dummy from 0 to 1) |                          |                     |                     |                        |                     |
|----------------------------------------|------------------------------------------------------------------------|--------------------------|---------------------|---------------------|------------------------|---------------------|
|                                        | No FE                                                                  | Burger King<br>County FE | District FE         | No FE               | McDonalds<br>County FE | District FE         |
| Own number of stores at t – 1          |                                                                        |                          |                     |                     |                        |                     |
| Dummy: Own #stores = 1                 | –0.021**<br>(0.005)                                                    | –0.036**<br>(0.007)      | –0.885**<br>(0.063) | –0.035**<br>(0.010) | –0.045**<br>(0.012)    | –0.550**<br>(0.056) |
| Dummy: Own #stores = 2                 | –0.023**<br>(0.004)                                                    | –0.030**<br>(0.005)      | –0.210*<br>(0.085)  | –0.047**<br>(0.006) | –0.060*<br>(0.008)     | –0.757**<br>(0.041) |
| Dummy: Own #stores ≥ 3                 | –0.019**<br>(0.005)                                                    | –0.027**<br>(0.005)      | –0.056<br>(0.036)   | –0.043**<br>(0.006) | –0.053**<br>(0.008)    | –0.816**<br>(0.038) |
| Competitor's number of stores at t – 1 |                                                                        |                          |                     |                     |                        |                     |
| Dummy: Comp.'s #stores = 1             | 0.032**<br>(0.011)                                                     | 0.037*<br>(0.014)        | –0.025<br>(0.055)   | 0.020<br>(0.013)    | 0.032*<br>(0.018)      | 0.052**<br>(0.073)  |
| Dummy: Comp.'s #stores = 2             | 0.045*<br>(0.023)                                                      | 0.052*<br>(0.029)        | –0.017<br>(0.031)   | 0.041<br>(0.029)    | 0.076<br>(0.046)       | –0.007**<br>(0.093) |
| Dummy: Comp.'s #stores ≥ 3             | 0.089**<br>(0.048)                                                     | 0.101*<br>(0.059)        | 0.011<br>(0.084)    | –0.041**<br>(0.007) | –0.050**<br>(0.009)    | –0.104**<br>(0.020) |
| Pred. Prob. Y = 1 at mean X            | 0.024                                                                  | 0.027                    | 0.014               | 0.045               | 0.054                  | 0.085               |
| Time dummies                           | YES                                                                    | YES                      | YES                 | YES                 | YES                    | YES                 |
| Control variables <sup>2</sup>         | YES                                                                    | YES                      | YES                 | YES                 | YES                    | YES                 |
| County Fixed Effects                   | NO                                                                     | YES                      | NO                  | NO                  | YES                    | NO                  |
| District Fixed Effects                 | NO                                                                     | NO                       | YES                 | NO                  | NO                     | YES                 |
| Number of Observations <sup>3</sup>    | 2110                                                                   | 1715                     | 535                 | 2110                | 1855                   | 640                 |
| Number of Local Districts <sup>3</sup> | 422                                                                    | 343                      | 107                 | 422                 | 371                    | 128                 |
| Log likelihood                         | –371.89                                                                | –340.26                  | –110.54             | –467.46             | –449.02                | –198.50             |
| Pseudo R-square                        | 0.229                                                                  | 0.252                    | 0.624               | 0.159               | 0.161                  | 0.441               |

Note 1: Estimated marginal effects are evaluated at the mean value of the rest of the explanatory variables. Note 2: Every estimation includes as control variables log-population, log-GDP per capita, log-population density, share population 5–14, share population 15–29, average rent, and proportion of claimants of unemployment benefits. Note 3: FE estimations do not include districts where the dependent variable does not have enough time variation. Note 4: \* and \*\* denote significance at the 5% and 1% level, respectively.

### Read:

- Table 7 shows that opening stores is effectively irreversible during the sample.
- The industry is still in an expansion phase: **41.6%** of market-year observations have no stores at all.
- States where BK is the local leader are less persistent; MD often expands into those markets.
- Table 8 is the motivating reduced-form fact: own lagged stores strongly reduce the probability of opening another store, but the rival's lagged stores have weak or even positive effects, especially for Burger King.
- Taken literally, that pattern is hard to reconcile with a standard static competition story, which is why the paper explores dynamic effects and biased beliefs.

TABLE 9  
Estimation of dynamic game for McDonalds and Burger King models with unbiased and biased beliefs<sup>(1)</sup> (Data: 422 markets, 2 firms, 5 years = 4,220 observations)

|                                         | $\beta = 0.95$ (not estimated) |                     |                           |                      |                           |                     |
|-----------------------------------------|--------------------------------|---------------------|---------------------------|----------------------|---------------------------|---------------------|
|                                         | (1)                            | (2)                 | (3)                       | (4)                  | (5)                       | (6)                 |
|                                         | Unbiased Beliefs               |                     | Biased Beliefs: $Z^* = 2$ |                      | Biased Beliefs: $Z^* = 1$ |                     |
|                                         | <i>BurgerKing</i>              | <i>McDonalds</i>    | <i>BurgerKing</i>         | <i>McDonalds</i>     | <i>BurgerKing</i>         | <i>McDonalds</i>    |
| Variable profits:                       |                                |                     |                           |                      |                           |                     |
| $\theta_0^{VP}$                         | 0.5413<br>(0.1265)*            | 0.8632<br>(0.2284)* | 0.4017<br>(0.2515)*       | 0.8271<br>(0.4278)*  | 0.4342<br>(0.2820)        | 0.8582<br>(0.4375)  |
| $\theta_{can}^{VP}$ cannibalization     | -0.2246<br>(0.0576)*           | 0.0705<br>(0.0304)* | -0.2062<br>(0.1014)*      | 0.0646<br>(0.0710)   | -0.1926<br>(0.1140)*      | 0.0640<br>(0.0972)  |
| $\theta_{com}^{VP}$ competition         | -0.0541<br>(0.0226)*           | -0.0876<br>(0.0272) | -0.1133<br>(0.0540)*      | -0.0856<br>(0.0570)  | -0.1381<br>(0.0689)*      | -0.0887<br>(0.0622) |
| Fixed costs:                            |                                |                     |                           |                      |                           |                     |
| $\theta_0^{FC}$ fixed                   | 0.0350<br>(0.0220)             | 0.0374<br>(0.0265)  | 0.0423<br>(0.0478)        | 0.0307<br>(0.0489)   | 0.0490<br>(0.0585)        | 0.0339<br>(0.0658)  |
| $\theta_{lin}^{FC}$ linear              | 0.0687<br>(0.0259)*            | 0.0377<br>(0.0181)* | 0.0829<br>(0.0526)*       | 0.0467<br>(0.0291)   | 0.0878<br>(0.0665)        | 0.0473<br>(0.0344)  |
| $\theta_{quad}^{FC}$ quadratic          | -0.0057<br>(0.0061)            | 0.0001<br>(0.0163)  | -0.0007<br>(0.0186)       | 0.0002<br>(0.0198)   | -0.0004<br>(0.0253)       | 0.0004<br>(0.0246)  |
| Entry cost:                             |                                |                     |                           |                      |                           |                     |
| $\theta_0^{EC}$ fixed                   | 0.2378<br>(0.0709)*            | 0.1887<br>(0.0679)* | 0.2586<br>(0.1282)*       | 0.1739<br>(0.0989)*  | 0.2422<br>(0.1504)        | 0.1764<br>(0.1031)  |
| $\theta_K^{EC}$ (K)                     | -0.0609<br>(0.043)             | -0.107<br>(0.0395)* | -0.0415<br>(0.096)        | -0.1190<br>(0.0628)* | -0.0419<br>(0.109)*       | -0.1271<br>(0.0762) |
| $\theta_Z^{EC}$ (Z)                     | 0.0881<br>(0.0368)*            | 0.0952<br>(0.0340)* | 0.1030<br>(0.0541)*       | 0.1180<br>(0.0654)*  | 0.0902<br>(0.0628)        | 0.1212<br>(0.0759)  |
| Log-Likelihood                          | -848.4                         |                     | -840.4                    |                      | -838.7                    |                     |
| Test of unbiased beliefs:               |                                |                     |                           |                      |                           |                     |
| For BK: $\hat{D}$ (d.o.f) ( $p$ -value) |                                |                     | 66.841 (32) (0.00029)     |                      | 66.841 (32) (0.00029)     |                     |
| For MD: $\hat{D}$ (d.o.f) ( $p$ -value) |                                |                     | 42.838 (32) (0.09549)     |                      | 42.838 (32) (0.09549)     |                     |

Note 1: Bootstrap standard errors in parentheses. Note 2: \* and \*\* denote significance at the 5% and 1% level, respectively.

## 5.6 Table 9: structural estimates with and without biased beliefs

Read:

- The test of unbiased beliefs clearly rejects equilibrium beliefs for Burger King:

$$LR = 66.841 \quad \text{with 32 d.f. and } p = 0.00029.$$

- For McDonalds, equilibrium beliefs are not rejected:

$$LR = 42.838 \quad \text{with 32 d.f. and } p = 0.09549.$$

- Once biased beliefs are allowed, the estimated competition effect of McDonalds on Burger King becomes much stronger in magnitude: the BK competition parameter more than doubles relative to the model that imposes unbiased beliefs.
- This is exactly the attenuation-bias story from the Monte Carlo section: if BK is over-optimistic about where MD will expand, then forcing unbiased beliefs makes BK look too profitable and competition look too weak.
- The paper also finds that unbiased beliefs would have raised BK profits in these markets by about **2.78%** in 1991, **2.11%** in 1992, **1.20%** in 1993, and **0.87%** in 1994.

## 6 Short summary

- Method.** The paper develops an identification and estimation framework for dynamic games in which players are rational but may have biased beliefs about rivals.
- Identification.** A payoff exclusion restriction generates a test of equilibrium beliefs and, with beliefs known at a subset of states, point identifies contemporaneous beliefs and payoff differences.
- Estimation.** The paper proposes a practical nonparametric two-step estimator and a parametric pseudo-likelihood extension.



- **Application.** In the UK store-location game between McDonalds and Burger King, the data reject unbiased beliefs for Burger King but not for McDonalds.
- **Main substantive result.** Imposing equilibrium beliefs when they are false attenuates estimated competition effects and can materially distort structural conclusions.

## 7 Conclusion

### 7.1 Core conclusion

- Dynamic games can be identified and estimated without imposing equilibrium beliefs everywhere.
- Standard payoff exclusion restrictions contain more information than is usually recognized: they do not just help identify payoffs; they also let the researcher test whether equilibrium beliefs are valid.
- A relatively weak anchor condition—known beliefs at a subset of states—is enough to recover contemporaneous beliefs and useful payoff objects.

### 7.2 Economic intuition

- The paper works because a player’s action reflects both payoffs and beliefs.
- If an excluded rival-specific state affects a player’s action only through beliefs, then movements in that state trace out the belief function.
- Once some points of the belief function are anchored, the rest of the structure can be recovered from best-response logic.
- Substantively, biased beliefs matter because they can masquerade as weak competition in reduced-form behavior.

### 7.3 Takeaway

This paper is important both methodologically and empirically. Methodologically, it shows how to test and estimate dynamic games when equilibrium beliefs are too strong an assumption. Empirically, it shows that allowing for biased beliefs changes what one learns about competitive interactions in a well-known store-location application. The broader lesson is that in strategic dynamic settings, researchers should treat equilibrium beliefs as a hypothesis to be tested, not a maintained assumption to be imposed automatically.